

① Sample format

Software Quality Assurance Plan

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An important part of achieving quality is to plan for quality, that is, to plan those activities that will help to achieve quality. The software quality assurance (SQA) plan is to provide a guideline for activities that will ensure the quality of software. I use the IEEE standard Std 730-1989 for developing the Software Quality Assurance Plan (SQAP) of my project.

1. Purpose

This SQA plan covers the software requirement, design, and implementation phases of the development of Internet Voting System Manager (IVSM). This plan does not cover the Cryptix and the Cryptix software that will be used by my project.

This SQAP provides a foundation for managing the IVSM software quality assurance activities, and is based on project activities and work products as documented in the project plan. This SQAP plan:

- Identifies the SQA responsibilities of the project developer and the SQA consultant
- Defines how to review and audit the development of IVSM system.
- Lists the activities, processes, and work products that the SQA consultant will review and audit
- Identifies the SQA work products

2. References

- Software Engineering, Roger S. Pressman
- Lecture Notes, CIS541 Software Engineering II, Dr. David Gustafson, Spring 2001.
- Lecture Notes, CIS740 Advanced Software Engineering, Dr. David Gustafson, Fall 1999.
- Lecture Notes, CIS748 Software Management, Dr. David Gustafson, Summer 1998.
- IEEE Standard for Software Test Documentation, IEEE Std 829-1983
- IEEE Standard for Software Quality Assurance Plans, ANSI/IEEE Std 730-1989
- IEEE Guide for Software Quality Assurance Planning, DRAFT, P730.2/D4

3. Management

Organization:

- The committee consists of Dr. Maria Zamfir-Bleyberg, Dr. William Hsu and Dr. Gustafson
- Major Professor: Dr. Maria Zamfir-Bleyberg
- Developer: MSE student Yonghua Li.

Tasks:

- Develop the requirement specification, cost estimation for distributed multiple sequence alignment system.
- Develop the design plan and test plan for distributed multiple sequence alignment system.
- Implement and test the distributed multiple sequence alignment system. Deliver the final version along with the documentation.
- On completion of the analysis, design and testing phases, the developer gives a formal presentation to the committee. The committee reviews the work performed by the developer and provides feedback.

Responsibilities

- The developer will perform all software development tasks for internet voting system manager under the supervision of the major professor.
- The committee will review the work performed by the developer and provide feedback and advice.

4. Documentation

The following documents will be provided at the end of each phase.

Phase 1: Software Requirement

- Project Overview
- System Diagram
- Project Gantt Chart Timeline
- Cost Analysis
- System Requirements Specification (SRS)

Phase 2: Software Design

- Software Quality Assurance Plan (SQAP)
- Object Model
- Formal Specifications
- Test Plan
- Formal Technical Review (checklist)

Phase 3: Software Implementation

- User Manual
- Testing and Reliable Evaluation

- Project Evaluation

Appendix:

- Source Code

5. Standard, Practices, Conventions, and Metrics

Standards

- Document Standards – MSE portfolio
- Coding Standards - Java 1.2
- Coding Documents - Java Documentation
- Test Standards - IEEE Standard for Software Test Documentation

Metrics

LOC - line of code is used to measure the size of the software

6. Review and Audits

There will be three formal presentations prepared by the developer and evaluated by the committee at the end of each phase (Requirement, Design, and Implementation). In the preliminary design review at the end of design phase, the technique adequacy of top-level design will be evaluated. All documents pertaining to one specific phase will be presented at the review. The evaluation is subject to be approved by the committee members. All deficiencies or inconsistency must be rectified by the developer and resubmitted to the committee members for approval.

7. Test, Tools, Techniques, and Methodologies

A Software Test Plan (STP) will be written to satisfy the software requirements. The plan will provide management and the testing function with an overview of the test activities, schedules and resources required to perform testing. The plan will describe how the testing specifications found in the following sections will be implemented.

Unit Test:

All code will be unit tested to ensure that the individual unit (class) performs the required functions and outputs the proper results and data.

Integration Test:

There are two levels of integration testing. One level is the process of testing a software capability. A second level of integration testing occurs when sufficient modules have been integrated to demonstrate a scenario.

Java Virtual Machine will be used for this project during the implementation. For measuring the size of the project, the function points analysis and COCOMO II were used. The design of the project is object oriented. Rational-Rose was used for making object diagram.

I will use the Cryptix framework to encrypt the communication between the server and client.

8. Problem Reporting and Corrective Action

During the whole development process of the project, the developer can report the problems to his major professor or his committee member. The major professor or committee member gives their suggestions. In the three presentations, the major professor or committee member also can ask questions and give their advice. After presentation, the developer corrects his mistakes.

9. Code control

N/A

10. Media control

N/A

11. Supplier control

The source code of the Cryptix and the Cryptix frameworks are from their developer's website. All these source code are reliable.

12. Records

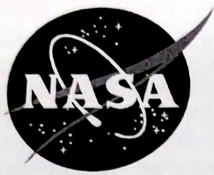
All the documents of each phase will be posted on my personal website in the CIS department. At the end of the phase III, the thesis composed of all the documents of the project will be committed to the CIS department.

13. Training

CIS540 Software Engineering Project
CIS740 Software Engineering
CIS748 Software Management

14. Risk Management

The most important risk is that the Cryptix framework is from third party. We do not know how well it works. I plan to run this system with boundary testing cases first, then apply it to general running.



<Project Name>

Software Quality Assurance (SQA) Plan

<Document Control Number>

Date: <Month Year>

CHECK THE <XXXXXX> AT< WEBSITE Address>

TO VERIFY THAT THIS IS THE CORRECT VERSION PRIOR TO USE.

General Tailoring Guidelines

This document is a template for a Software Quality Assurance Plan intended for use by Code 303, Assurance Management Office (AMO) personnel as the basis for a mission-specific Software Quality Assurance Plan.

There are three general types of text format: Text in Black, Text in Blue with < >, Text in Blue with [] and Text underlined in blue.

Text in Black

- Text in this style, black, is used for text that is equally applicable to all Software Quality Assurance Plans and will be included in the Software Quality Assurance Plan without modification. All document section headings are in the same category.

Text in Blue with < >

- Text in this style, <blue>, is used to indicate that the text needs to be updated with project specific information such as the project name, persons name, persons title, date, revision #, Document Control Number, exc...

Text in Blue with []

- Text in this style, [blue], is used to indicate that there is additional instruction for tailoring text in any specific section. Delete this style, [blue], text before the document is finalized.

Text underlined in Blue

- Text in this style, blue, is used to indicate active links. These links are usually to project or software assurance websites. The text color may vary depending on each persons personal computer settings/configuration.

All components of the table of contents will be addressed, but the level of detail is left up to the software quality personnel. The length and level of detail of the Software Quality Assurance Plan should be commensurate with the scope and complexity of the project.

Section headings may be added where necessary, but **existing headings should not be modified or deleted**. If a particular section is not applicable to the specific Software Quality Assurance Plan under production, that fact should be noted under the section heading, together with a brief explanation.

The following disclaimer appears on all pages: "Printed copies of this document are for REFERENCE PURPOSES ONLY! The only controlled copy of this document is located on-line at <http://xxxxxxx>. This disclaimer should be modified to contain the appropriate URL, but should not be removed.

Finally, in the Software Assurance Plan Template, this entire section ("**General Tailoring Guidelines**") will be deleted.

Foreword

This document is a <Project Name> Project controlled document and adheres to IEEE 730-2002, the IEEE Standard for Software Quality Assurance Plans. Changes to this document require prior approval of the <Project Name> Project Configuration Control Board (CCB). Proposed changes shall be submitted to the <Project Name> Systems Assurance Manager (SAM), along with supportive material justifying the proposed change.

Questions or comments concerning this document should be addressed to the Assurance Management Office:

<Project Name>, <SAM Name>/Code 303
Building <XX> Room <XXX>
Mail Stop <XXX>
Goddard Space Flight Center
Greenbelt, Maryland 20771
(301) <XXX-XXXX>

Signature Page

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1.0 Purpose

The purpose of this Software Quality Assurance (SQA) Plan is to establish the goals, processes, and responsibilities required to implement effective quality assurance functions for the <Project Name> project.

The <Project Name> Software Quality Assurance Plan provides the framework necessary to ensure a consistent approach to software quality assurance throughout the project life cycle. It defines the approach that will be used by the SAM and Software Quality (SQ) personnel to monitor and assess software development processes and products to provide objective insight into the maturity and quality of the software. The systematic monitoring of <Project Name> products, processes, and services will be evaluated to ensure they meet requirements and comply with National Aeronautics and Space Administration (NASA), Goddard Space Flight Center (GSFC), and <Project Name> policies, standards, and procedures, as well as applicable Institute of Electrical and Electronic Engineers (IEEE) standards.

1.1 Scope

This plan covers SQA activities throughout the <identify phases, e.g., formulation and implementation> phases of the <Project Name> mission. [Indicate whether or not SQA activities will continue through operations and maintenance of the system.]

[Enter a brief description of the project, the suppliers of the software, the software items covered by the plan, and their intended use.]